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Paper ID #54370

WIP: Training Prompt Engineering Skills in AI-Assisted Programmatic Labeling Tasks

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Abstract

Prompt engineering is a critical skill for effective human–AI collaboration, yet there is limited empirical research on how people learn to improve prompts for interacting with large language models (LLMs). This work-in-progress presents an interactive system and a mixed-methods study design to examine how different feedback strategies influence the development of prompt engineering skills in AI-assisted text labeling tasks. Participants iteratively write prompts to guide an LLM-based labeler using standard X/Twitter datasets, with the goal of improving labeling accuracy. The study compares three feedback conditions: (1) performance metrics only, (2) fixed, predefined feedback, and (3) adaptive AI-generated feedback. Quantitative measures track performance changes across iterations, while qualitative surveys capture participants' understanding, confidence, and perceived usefulness of feedback. By focusing on iterative refinement and feedback-driven learning, this work aims to inform pedagogical approaches to teaching prompt engineering as a practical human-in-the-loop skill. In addition to classification performance metrics, the analysis also examines prompt revision patterns and participant reflections to explore whether improvements reflect participants' own prompt engineering development or short-term reliance on AI-generated suggestions.

Keywords: AI education, Prompt engineering, human AI collaboration, human in the loop, AI-assisted labeling, feedback strategies, large language models, decision support systems, mixed methods study, interactive systems

Introduction

Large language models (LLMs) are increasingly integrated into data analysis and decision-support workflows, including text classification and annotation tasks [1, 2, 3, 4]. In many real-world applications, LLMs act as automated labelers that rely on natural language prompts to infer labeling rules and apply them consistently across datasets. Prior work has demonstrated that the structure and formatting of prompts can substantially influence LLM performance, even when task content remains unchanged [5]. As a result, effective prompt engineering has emerged as a critical component of successful human–AI collaboration.

Despite this sensitivity, users interacting with LLM-based classification systems vary widely in their ability to formulate prompts that clearly convey labeling intent. In practice, users often struggle to describe nuanced label definitions and edge cases, particularly when working with open-

ended, real-world data. These challenges underscore the need to better understand how prompt engineering skills develop through interaction with AI systems and how feedback can support users in refining their prompts over time. This work-in-progress presents an interactive system designed to study how different feedback strategies influence the development of prompt engineering skills in an AI-assisted text labeling task. Participants iteratively write prompts to guide an LLM-based labeler and receive one of three types of feedback: quantitative performance metrics, fixed guided feedback, or adaptive AI-generated feedback. By examining how these feedback strategies shape prompt refinement and learning, this study aims to inform the design of educational tools and pedagogical frameworks for teaching prompt engineering as a practical, human-in-the-loop skill.

Background and Motivation

Prior research has shown that prompt formulation plays a significant role in determining LLM output quality. Even with advanced models, variations in prompt structure, formatting, and instruction clarity can lead to substantial differences in performance [5]. These findings underscore the importance of prompt engineering, not only as a technical consideration, but as a user-facing skill that shapes the effectiveness of human–AI collaboration. Challenges in prompt formulation are particularly evident in text classification tasks involving social media data. Datasets derived from platforms such as X (formerly Twitter) are characterized by informal language, contextual ambiguity, and rapidly evolving discourse. Prior work in social media natural language processing highlights the difficulty of achieving reliable labeling under these conditions, even when ground-truth annotations are available [6]. As a result, users must often articulate nuanced, context-dependent labeling rules to effectively guide AI systems. Recent work has begun to frame prompt engineering as a learnable skill rather than a purely technical configuration task [7]. Studies examining prompt engineering instruction suggest that training can influence users' perceived expertise, confidence, and interaction outcomes, though learning effects vary depending on task complexity and prior knowledge [8]. Educational research further emphasizes that learners bring diverse backgrounds and levels of familiarity with AI systems, making one-size-fits-all instructional approaches insufficient [9]. Notably, much of the existing work relies on static, pre-task instruction rather than feedback that adapts to users' behavior during task execution. This study is motivated by the potential of feedback-driven learning to address these limitations. Rather than assuming a single effective feedback approach, this work-in-progress explicitly compares multiple feedback strategies within the system design. In particular, it explores whether adaptive, AI-generated feedback capable of analyzing a user's prompt, task performance, and prompting history can offer different learning affordances than quantitative metrics alone or fixed guided feedback. At the same time, the study acknowledges important open questions, including the risk of over-reliance on AI-generated guidance, the role of prior AI knowledge in shaping learning outcomes, and whether explicit instruction in prompting techniques remains necessary as LLMs continue to advance.

System and Study Design

System Overview The system is designed to support learning prompt engineering through an AI-assisted text classification task. Participants are assigned a dataset of 100 short X/Twitter posts, labeled into two categories: humanitarian and non-humanitarian. Humanitarian tweets include

content relevant to disaster response, such as safety warnings, requests for assistance, or reports of damage, while non-humanitarian tweets consist of unrelated commentary, opinions, or general statements. The dataset is derived from the CrisisMMD corpus, a publicly available, human-annotated collection of crisis-related social media posts collected during multiple natural disasters [10].

Participants complete the labeling task through an iterative workflow in which they write a prompt, run the labeling process, review the resulting performance metrics, and revise the prompt in subsequent attempts. In addition to recording model performance metrics, the system also stores the sequence of prompt revisions submitted by each participant, enabling analysis of how prompts evolve across iterations during the labeling task.

Prompting Frameworks To introduce participants to common prompt engineering strategies [11], the system allows users to select one of four prompting frameworks before each attempt: zero-shot, few-shot, chain-of-thought, or structured reasoning. These frameworks function as instructional scaffolds rather than model-level configurations; the underlying LLM and evaluation procedure remain constant across all conditions. Zero-shot prompting encourages participants to describe the task and label definitions without providing examples. Few-shot prompting guides participants to include representative input–output examples within the prompt. Chain-of-thought prompting encourages participants to request explicit decision criteria or reasoning steps for classification. Structured reasoning prompts participants to organize instructions using clearly defined sections, such as task description, label definitions, and decision rules. Participants may revise their prompts within the same framework across attempts, allowing exploration of how refinement within a given strategy influences performance.

Feedback Conditions The study compares three feedback conditions that differ in the type of guidance participants receive after each prompt attempt. Figure 1 illustrates how feedback is incorporated into the iterative prompt engineering workflow.

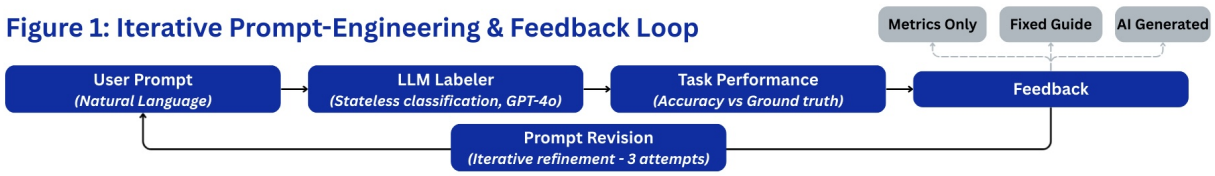


Figure 1: Feedback conditions in the study. Participants iteratively refine prompts while receiving performance metrics only, fixed guided feedback, or adaptive AI-generated feedback.

In all conditions, participants interact with the same labeling interface and dataset. After submitting a prompt, the system runs the LLM-based classifier and returns performance metrics comparing model predictions with ground-truth labels. Participants then revise their prompts based on the feedback received.

The first condition provides **performance metrics only**. Participants are shown quantitative indicators of labeling performance, including accuracy, precision, recall, and F1-score. In this con-

dition, participants must independently interpret the results and determine how to modify their prompts.

The second condition provides **fixed guided feedback**. In addition to performance metrics, participants receive a predefined paragraph of general prompt engineering guidelines. This guidance includes strategies such as clarifying label definitions, specifying classification criteria, and providing examples. The same guidance text is presented to all participants regardless of their prompt content.

The third condition provides **adaptive AI-generated feedback**. In this condition, an instructed LLM analyzes the participant's prompt and task performance to generate targeted suggestions for improvement. The feedback may highlight ambiguities in label definitions, suggest prompt restructuring, or recommend additional examples.

Across all conditions, the labeling task, dataset, and underlying LLM classifier remain identical. Only the form of feedback differs, allowing the study to examine how different feedback mechanisms influence prompt refinement behavior.

Measures and Planned Analysis

Quantitative Measures Participants' prompt engineering performance is evaluated using classification metrics computed after each prompt submission. Specifically, the system compares the LLM's predicted labels with the ground-truth dataset to calculate accuracy, precision, recall, and macro-averaged F1 score. These metrics allow the study to measure how labeling performance changes across prompt iterations as participants refine their prompts.

In addition to task performance, the study analyzes the sequence of prompt revisions submitted by each participant. To explore whether improvements result from participants' own prompt refinement or direct adoption of AI suggestions, the study compares the similarity between AI-generated feedback suggestions and participants' subsequent prompt revisions. This comparison provides an initial estimate of the extent to which users incorporate AI feedback directly versus independently modifying their prompts during the iterative labeling process.

Qualitative Measures In addition to performance metrics, the study collects qualitative data to capture participants' experiences with the prompt engineering process and their perceptions of the feedback received. After completing the task, participants complete a short survey and reflection questions describing their prompting strategies, perceived difficulty of the labeling task, and perceived usefulness of the feedback provided.

The survey also asks participants to report their prior experience with large language models and their perceptions of generative AI tools before and after the study. To better understand potential reliance on AI assistance, participants are asked to rate how much they depended on AI-generated feedback during prompt revision and whether they believe they could apply similar prompting strategies in future tasks without assistance. Responses to these questions provide additional context for interpreting the quantitative results and help identify whether performance improvements reflect prompt engineering learning or short-term reliance on AI suggestions.

Discussion, Limitations, and Future Work

This work-in-progress explores how different feedback strategies may support the development of prompt engineering skills in AI-assisted text classification tasks. By focusing on iterative interaction rather than one-time instruction, the study positions prompt engineering as a learning process shaped by feedback and reflection.

Several limitations should be acknowledged. The current study focuses on binary classification tasks using social media data, which may limit generalizability to other domains or more complex labeling scenarios. Participant samples consist of undergraduate students from a single institution, and learning outcomes may differ for users with greater domain expertise or professional experience. Additionally, although the study analyzes prompt revisions and participant reflections, the short duration of the experiment limits the ability to fully distinguish long-term prompt engineering learning from short-term reliance on AI-generated suggestions.

Future work will explore extending the system to support user-uploaded datasets, multi-class classification tasks, and longitudinal study designs to examine retention and transfer of prompt engineering skills. Further investigation is also needed to understand how users' prior knowledge, perceptions of AI systems, and trust in automated feedback influence learning outcomes.

Conclusion

This work-in-progress presents an interactive system and study design for examining how feedback strategies influence the development of prompt engineering skills in AI-assisted text classification tasks. By comparing quantitative metrics, fixed guided feedback, and adaptive AI-generated feedback within a controlled, iterative setting, the study aims to better understand how users refine prompts through interaction and feedback. While limited in scope, this work contributes to ongoing discussions in AI education and human–AI collaboration by highlighting the role of feedback in developing practical prompt-engineering skills. The findings from this study are intended to inform future pedagogical frameworks for teaching prompt engineering as a human-in-the-loop competency.

IRB Approval

This study was reviewed and approved by the Institutional Review Board under protocol number IRB-2037012-3.

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